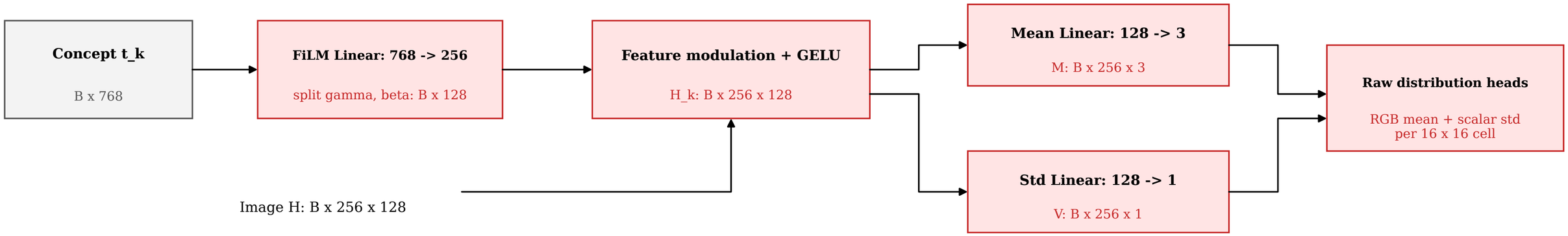
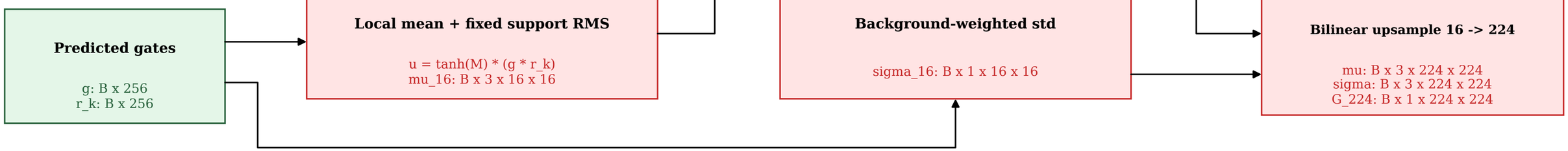
Inside the Concept-Conditional RGB Noise Generator

(a) Condition the patch features on the selected concept text



$$H_k = \mathrm{GELU}(H \odot (1+\gamma_k) + \beta_k)$$

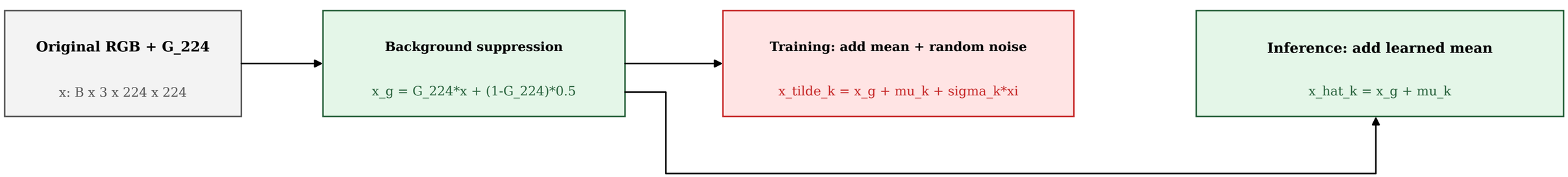
(b) Put mean shift on concept support; protect foreground from high variance



$$\mu_{16} = b_\mu u / \sqrt{\sum u^2 / (3 \sum (g r_k)^2)}; \quad b_\mu = 0.02686$$

$$\sigma_{16} = 0.008 + (1-g)(0.08 + 0.12\,\mathrm{sigmoid}(V)) + 0.025 g (1-r_k)$$

(c) Sample RGB in training; keep only the learned mean during inference



$$\xi \sim \mathcal{N}(0, I), \quad \xi, \tilde{x}_k, \hat{x}_k \in \mathbb{R}^{B \times 3 \times 224 \times 224}$$